

(1)

1. List all distinct departments in the student table	<pre>SELECT DISTINCT department FROM students;</pre>	<table><tr><th>department</th></tr><tr><td>IT</td></tr><tr><td>HR</td></tr><tr><td>Finance</td></tr></table>	department	IT	HR	Finance								
department														
IT														
HR														
Finance														
2. Get the average age of students per department	<pre>SELECT department, AVG(age) AS avg-age FROM students GROUP BY department;</pre>	<table><tr><th>department</th><th>avg-age</th></tr><tr><td>IT</td><td>20.5</td></tr><tr><td>HR</td><td>22</td></tr><tr><td>Finance</td><td>23</td></tr></table>	department	avg-age	IT	20.5	HR	22	Finance	23				
department	avg-age													
IT	20.5													
HR	22													
Finance	23													
3. Show departments with more than 1 student	<pre>SELECT department, COUNT(student-id) AS student-count FROM students GROUP BY department HAVING student-count > 1;</pre>	<table><tr><th>department</th><th>student-count</th></tr><tr><td>IT</td><td>2</td></tr><tr><td>HR</td><td>2</td></tr></table>	department	student-count	IT	2	HR	2						
department	student-count													
IT	2													
HR	2													
4. Get all students whose age is between 21 and 23	<pre>SELECT student-id, name, age, department FROM students WHERE age BETWEEN 21 AND 23</pre>	<table><tr><th>student-id</th><th>name</th><th>age</th><th>department</th></tr><tr><td>2</td><td>bob</td><td>22</td><td>HR</td></tr><tr><td>5</td><td>Eve</td><td>22</td><td>HR</td></tr></table>	student-id	name	age	department	2	bob	22	HR	5	Eve	22	HR
student-id	name	age	department											
2	bob	22	HR											
5	Eve	22	HR											

Created: Wed, 4/29/2026



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5. List all students in the IT or HR department who are older than 21	<pre>SELECT student_id, name, age, department FROM students WHERE department = 'IT' OR 'HR' AND age > 21 ;</pre>	<table><thead><tr><th>student-id</th><th>name</th><th>age</th><th>department</th></tr></thead><tbody><tr><td>2</td><td>Bob</td><td>22</td><td>HR</td></tr><tr><td>5</td><td>Eve</td><td>22</td><td>HR</td></tr></tbody></table>	student-id	name	age	department	2	Bob	22	HR	5	Eve	22	HR				
student-id	name	age	department															
2	Bob	22	HR															
5	Eve	22	HR															
6. Show total credits per department, only for department with more than 5 total credits.	<pre>SELECT department sum(credits) AS total_credits FROM courses GROUP BY department HAVING total_credits > 5 ;</pre>	<table><thead><tr><th>department</th><th>total-credits</th></tr></thead><tbody><tr><td>IT</td><td>11</td></tr></tbody></table>	department	total-credits	IT	11												
department	total-credits																	
IT	11																	
7. List all courses that do not have 4 credits	<pre>SELECT course_id, course_name, department, credits FROM courses WHERE credits != 4 ;</pre>	<table><thead><tr><th>course-id</th><th>course-name</th><th>department</th><th>credits</th></tr></thead><tbody><tr><td>101</td><td>SAL Basics</td><td>IT</td><td>3</td></tr><tr><td>104</td><td>Excel</td><td>Finance</td><td>2</td></tr><tr><td>105</td><td>Statistics</td><td>HR</td><td>3</td></tr></tbody></table>	course-id	course-name	department	credits	101	SAL Basics	IT	3	104	Excel	Finance	2	105	Statistics	HR	3
course-id	course-name	department	credits															
101	SAL Basics	IT	3															
104	Excel	Finance	2															
105	Statistics	HR	3															
8. Show the top 3 courses by credits in descending order	<pre>SELECT course_id, course_name, credits FROM courses ORDER BY credits DESC LIMIT 3 ;</pre>	<table><thead><tr><th>course-id</th><th>course-name</th><th>credits</th></tr></thead><tbody><tr><td>102</td><td>Python</td><td>4</td></tr><tr><td>103</td><td>Data Science</td><td>4</td></tr><tr><td>101</td><td>SAL Basics</td><td>3</td></tr></tbody></table>	course-id	course-name	credits	102	Python	4	103	Data Science	4	101	SAL Basics	3				
course-id	course-name	credits																
102	Python	4																
103	Data Science	4																
101	SAL Basics	3																

Created: Wed, 4/29/2026



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9. Get the maximum minimum and average grade across all enrollments	<pre>SELECT max(grade) AS max_grade, min(grade) AS min_grade, avg(grade) AS avg_grade FROM enrollments</pre>	<table><tr><th>max_grade</th><th>min_grade</th><th>avg_grade</th></tr><tr><td>90</td><td>78</td><td>84.6</td></tr></table>	max_grade	min_grade	avg_grade	90	78	84.6						
max_grade	min_grade	avg_grade												
90	78	84.6												
10. Count how many enrollments exist per course	<pre>SELECT COUNT(enrollment_id) AS enrollment_count, course_id FROM enrollments GROUP BY course_id;</pre>	<table><tr><th>Course_id</th><th>enrollment_id</th></tr><tr><td>101</td><td>1</td></tr><tr><td>102</td><td>1</td></tr><tr><td>103</td><td>1</td></tr><tr><td>104</td><td>1</td></tr><tr><td>105</td><td>1</td></tr></table>	Course_id	enrollment_id	101	1	102	1	103	1	104	1	105	1
Course_id	enrollment_id													
101	1													
102	1													
103	1													
104	1													
105	1													
11. Find total salary and total bonus per department.	<pre>SELECT SUM(salary) AS total_salary, SUM(bonus) AS total_bonus, department FROM salaries GROUP BY department;</pre>	<table><tr><th>department</th><th>total_salary</th><th>total_bonus</th></tr><tr><td>IT</td><td>122 000</td><td>10500</td></tr><tr><td>HR</td><td>109 000</td><td>7500</td></tr><tr><td>Finance</td><td>70 000</td><td>6000</td></tr></table>	department	total_salary	total_bonus	IT	122 000	10500	HR	109 000	7500	Finance	70 000	6000
department	total_salary	total_bonus												
IT	122 000	10500												
HR	109 000	7500												
Finance	70 000	6000												
12. Show departments where average salary is above 55 000	<pre>SELECT department, AVG(salary) AS avg_salary FROM salaries HAVING avg_salary > 55 000;</pre>	<table><tr><th>department</th><th>avg_salary</th></tr><tr><td>IT</td><td>61 000</td></tr><tr><td>Finance</td><td>70 000</td></tr></table>	department	avg_salary	IT	61 000	Finance	70 000						
department	avg_salary													
IT	61 000													
Finance	70 000													

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13. List employees whose salary plus bonus is greater than 60000	<pre>SELECT employee_id, name, salary, bonus, bonus + salary AS total_compensation FROM salaries HAVING total_compensation > 60000</pre>	<table border="1"> <thead> <tr> <th>Employee_id</th> <th>name</th> <th>salary</th> <th>bonus</th> <th>Total-compensation</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>Tam</td> <td>60000</td> <td>5000</td> <td>65000</td> </tr> <tr> <td>3</td> <td>Spina</td> <td>7000</td> <td>6000</td> <td>13000</td> </tr> <tr> <td>4</td> <td>Tyke</td> <td>62000</td> <td>5000</td> <td>67000</td> </tr> </tbody> </table>	Employee_id	name	salary	bonus	Total-compensation	1	Tam	60000	5000	65000	3	Spina	7000	6000	13000	4	Tyke	62000	5000	67000
Employee_id	name	salary	bonus	Total-compensation																		
1	Tam	60000	5000	65000																		
3	Spina	7000	6000	13000																		
4	Tyke	62000	5000	67000																		
14. Show total and average budget per department. only include departments with average budget above 70000	<pre>SELECT department, sum(budget) AS total-budget, avg(budget) AS avg-budget FROM projects HAVING avg-budget > 70000;</pre>	<table border="1"> <thead> <tr> <th>department</th> <th>total-budget</th> <th>avg-budget</th> </tr> </thead> <tbody> <tr> <td>IT</td> <td>270000</td> <td>135000</td> </tr> <tr> <td>Finance</td> <td>80000</td> <td>40000</td> </tr> </tbody> </table>	department	total-budget	avg-budget	IT	270000	135000	Finance	80000	40000											
department	total-budget	avg-budget																				
IT	270000	135000																				
Finance	80000	40000																				
15. List all projects with budgets between 50000 and 120000 excluding the Marketing department.	<pre>SELECT project_id, project_name, department, budget FROM projects WHERE department != 'Marketing' AND budget BETWEEN 50000 and 120000</pre>	<table border="1"> <thead> <tr> <th>project_id</th> <th>project_name</th> <th>department</th> <th>budget</th> </tr> </thead> <tbody> <tr> <td>2</td> <td>payroll system</td> <td>Finance</td> <td>80000</td> </tr> </tbody> </table>	project_id	project_name	department	budget	2	payroll system	Finance	80000												
project_id	project_name	department	budget																			
2	payroll system	Finance	80000																			

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